Lab 00 · Exercise 1

Goal: reproduce the following display in a Colab notebook.

Exercise: Some simple Statistics tasks

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1. Arithmetic Mean
2. Mean Squared Error

   A. Bias of an estimator
   B. Variance of an estimator

1. The Arithmetic Mean

$$Mean(x) = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Embedded a formatted code block:

```
x=[1,3,9,4]
x_bar=1/4*sum(x)
```

```
[2]  # actual python calculation
     x=[1,3,9,4]
     x_bar=1/len(x)*sum(x)
     x_bar
```

```
4.25
```

2. Mean Squared Error

$$MSE(x) = bias^2(x) + var(x)$$

For further information see http://www.statisticshowto.com/mean-squared-error